

Depth profiling of GaN by cathodoluminescence microanalysis

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We present the results of a depth-resolved cathodoluminescence (CL) and transmission electron microscopy study of autodoped GaN grown on sapphire. Depth-resolved CL analysis can be used for depth profiling of the yellow luminescence (YL) center concentration which was found to increase with depth. The results are consistent with the $(\text{O}_\text{N}-\text{V}_\text{Ga})^{2-}$ complex model of YL centers [J. Neugebauer and C. G. Van de Walle, *Appl. Phys. Lett.* **69**, 503 (1996) and T. Mattila and R. M. Nieminen, *Phys. Rev. B* **55**, 9571 (1996)]. Depth profiling of the near-edge emission in GaN layers thicker than $\sim 0.5 \mu\text{m}$ is not possible due to strong self-absorption. © 1999 American Institute of Physics. [S0003-6951(99)04108-X]

Research in GaN has recently intensified due to its favorable properties for short-wavelength optoelectronic devices.¹ Defect-induced, broad emissions such as yellow luminescence (YL) and the as yet unexplained blue luminescence (BL) observed in *n*-type GaN can play significant roles in the luminescent properties of bulk and epitaxial GaN.^{2,3} YL, the most frequently reported defect-induced emission in autodoped GaN, has been attributed to a shallow donor–deep acceptor transition.⁴ The deep acceptor involved in YL is believed to be the native gallium vacancy (V_Ga^{3-}) defect.⁵ The shallow donor states involved in YL and the *n*-type character of autodoped GaN have been attributed to substitutional oxygen (O_N^+) and silicon (Si_Ga^+) impurities.^{2,3} Depth-resolved secondary ion mass spectrometry (SIMS) studies of epitaxial GaN have revealed a strong dependence of impurity concentrations on the distance from the substrate. Concentrations of O and Si rapidly increase in approximately the first micron of films grown on Al_2O_3 and SiC substrates, respectively. The increase has been attributed to impurity diffusion from substrates during growth.^{6,7} In a depth-resolved cathodoluminescence (CL) study, Li and Coleman⁸ have reported an exponential decrease in the ratio of the band edge to YL emission intensity with depth, attributed to changes in epilayer quality. The results presented here also illustrate an increase in YL intensity with depth, but the intensity of the near-edge emission as a function of depth is shown to be dominated by self-absorption.⁹ The presented CL and transmission electron microscopy (TEM) results are consistent with reported studies of defect and impurity distributions and models of the origin of YL.

The sample used in this study was an unintentionally doped 4 μm thick epilayer of wurtzite GaN grown at 1040 °C on a 25 nm GaN buffer layer by metal–organic chemical vapor deposition. The buffer was grown at 495 °C

on a *c*-plane sapphire substrate. Trimethylgallium and ammonia were used as precursors. Room-temperature CL and wavelength-dispersive x-ray spectroscopy (WDS) measurements were performed using an Oxford Instruments MonoCL2 CL system and a Microspec WDS x-ray system both installed on a JEOL35C scanning electron microscope (SEM) equipped with a liquid-helium cold stage. The CL signal was dispersed by a 1200 lines/mm grating blazed at 500 nm and detected using a Hamamatsu R943-02 peltier-cooled photomultiplier tube. The electron-beam current (I_b) was measured using a Faraday cup. TEM measurements were performed in a Philips 430 operating at 300 keV.

CL generation throughout the electron interaction volume is governed by the local density of electron–hole (e–h) pairs. The e–h pair generation profile and the CL generation profile within the interaction volume can be approximated by the total primary electron energy-loss profile.¹⁰ The energy-loss depth profile maximizes at approximately $0.3R_\text{KO}$,¹⁰ where R_KO is the Kanaya–Okayama electron penetration range.¹¹ Depth-resolved CL information can be obtained by measuring CL intensity as a function of R_KO . R_KO is related to the electron-beam energy (E_b) by the Kanaya–Okayama equation.¹¹ Depth-resolved CL experiments are generally conducted under constant beam power ($E_b I_b$) or constant power density ($E_b I_b / R_\text{KO}$) conditions since the e–h pair generation rate in the interaction volume is proportional to $E_b I_b$.¹⁰ In this study we adopted the former approach. Changes in the signal intensity in CL depth profiles are caused by variations in physical properties throughout the sample (which can be induced by the electron beam), self-absorption, surface recombination, saturation of extrinsic CL sources, and recombination competition effects.¹²

A room-temperature CL spectrum, corrected for system response, obtained using a beam energy of 35 keV and a beam current of 1 nA is shown in Fig. 1. The spectrum consists of a near-edge GaN emission at 3.38 eV, the BL and YL centered on approximately 2.8 and 2.2 eV, respectively, and the $\text{Cr}_\text{Al}^0\text{:Al}_2\text{O}_3$ emission from the substrate at 1.79 eV.

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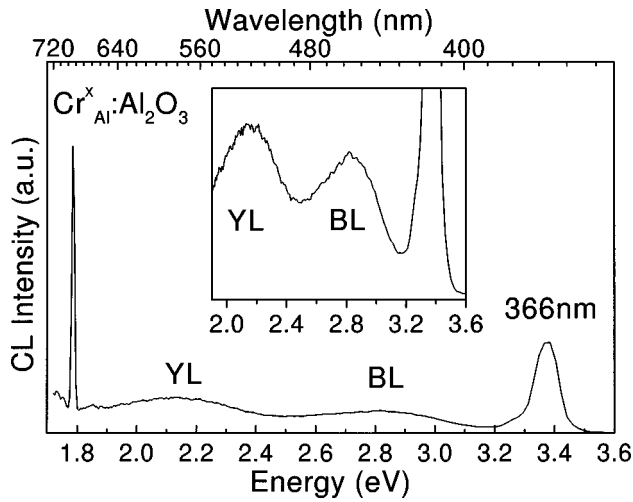


FIG. 1. Room-temperature CL spectrum of autodoped GaN obtained using a beam energy of 35 keV, beam current of 1 nA, and corrected for system response. The inset illustrates the (corrected) shapes of the YL and BL peaks.

The ripples visible in the BL and YL bands are due to the microcavity effect.¹³ The emissions were observed to vary with irradiation time under high-dose conditions due to electron-beam-induced impurity electromigration.¹⁴ Depth-resolved CL measurements were, therefore, performed using sufficiently low-beam currents ($E_b I_b = 14 \mu\text{W}$) and SEM magnification ($\times 1000$) to ensure a negligible contribution to CL intensity modulation by electromigration and saturation effects. Depth-resolved CL data were obtained by measuring the average CL intensity of the near-edge, BL, YL, and the 1.79 eV emissions produced by 10 s irradiation using beam energies in the range 5–35 keV. E_b was stepped at 2 keV intervals. I_b was adjusted prior to each measurement to maintain constant beam power. Spectra were not acquired at each E_b to minimize the total irradiation time and, hence, the net (additive) effects of electromigration. Spectral resolution of 10 nm was used to account for the shift of the near-edge emission with E_b due to self-absorption.⁹

Total primary electron energy-loss profiles were calculated using the Monte Carlo simulation CASINO documented in Ref. 15. To a first approximation, the energy loss is proportional to the e–h pair generation rate.¹⁰ The energy-loss profiles were obtained by adding a routine to CASINO that divides the interaction volume into 500 slices, traces every simulated electron trajectory, and calculates the energy deposited into each slice.¹⁰ At each E_b , 10^5 electron trajectories were simulated. The constant beam power condition was simulated by dividing every point on each energy-loss profile by E_b . Since the measured near-edge emission intensity is strongly affected by self-absorption,⁹ the CASINO CL generation profiles were corrected using the Beer–Lambert Law:

$$I(z) = I_0(z) e^{-\alpha z}, \quad (1)$$

where $I(z)$ is the (modeled) detected intensity of the near-edge signal generated at depth z , $I_0(z)$ is the electron energy loss at z , and α is the absorption coefficient. The absorption coefficient decays exponentially across the low-energy tail of the near-edge emission.⁹ Accordingly, the values $\alpha = 0.63 \times 10^5$ and $0.1 \times 10^5 \text{ cm}^{-1}$ were used, respectively, for 80% and 20% of the (simulated) signal generated at each depth.

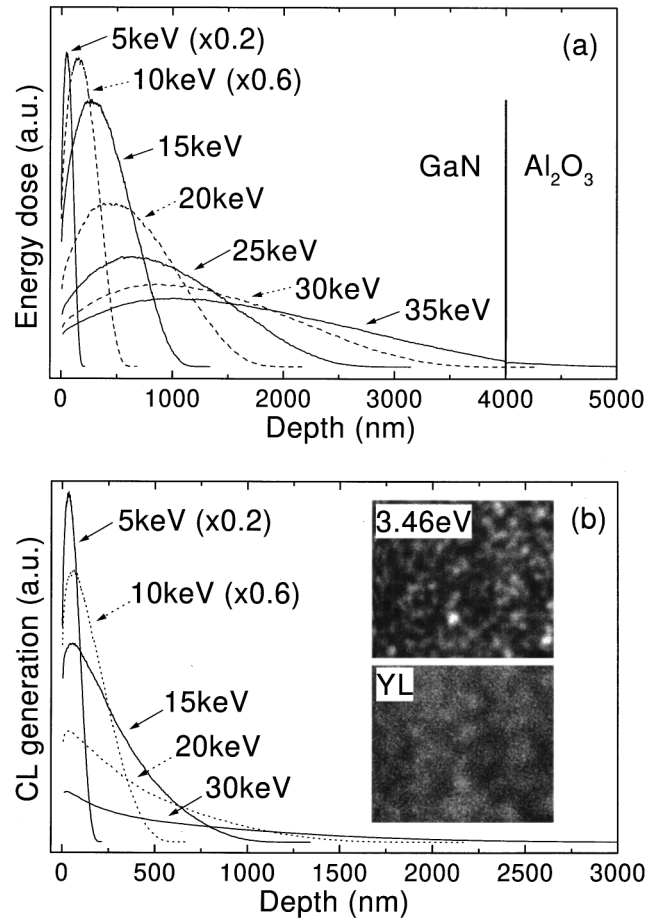


FIG. 2. (a) Simulated total energy-dose profiles for a $4 \mu\text{m}$ GaN epilayer grown on a sapphire substrate calculated using a number of beam energies. The profiles were used to approximate the YL and BL CL generation profiles. (b) Simulated generation profiles for the measured near-edge CL signal obtained by correcting the profiles in (a) for self-absorption using Eq. (1). The inset shows CL images obtained at 4 K using the near-edge and YL emissions ($E_b = 15 \text{ keV}$, field of view = $20 \mu\text{m}$).

The energy-loss profiles calculated for a $4 \mu\text{m}$ thick epilayer of GaN (density = 6.11 g cm^{-3}) on sapphire (density = 3.9 g cm^{-3}) are shown in Fig. 2(a) for a number of beam energies (constant beam power). Profiles corrected for self-absorption using Eq. (1) are shown in Fig. 2(b). The near-edge emission depth profile was simulated by integrating the CL generation curves corrected for self-absorption [Fig. 2(b)] within the GaN epilayer (first $4 \mu\text{m}$). As self-absorption of the YL and BL emissions is negligible⁹ and CL generation is proportional to $E_b I_b$,¹⁰ the YL and BL CL depth profiles for a homogeneous epilayer are constant under constant beam power conditions. The profiles were simulated by integrating the energy-dose profiles in Fig. 2(a) over the first $4 \mu\text{m}$. The simulated and measured depth profiles are shown in Fig. 3. Also shown in Fig. 3 are the $\text{Cr}_{\text{Al}}^0\text{:Al}_2\text{O}_3$ simulated CL depth profiles, obtained by integrating the energy-loss profile within the substrate, and the measured $\text{Cr}_{\text{Al}}^0\text{:Al}_2\text{O}_3$ depth profile. The simulated profiles in Fig. 3 are arbitrarily scaled since they only reflect the profile shape as a function of E_b . As expected, the simulated $\text{Cr}_{\text{Al}}^0\text{:Al}_2\text{O}_3$ intensity starts to increase when R_{KO} exceeds the GaN epilayer thickness. The measured $\text{Cr}_{\text{Al}}^0\text{:Al}_2\text{O}_3$ profile increases with E_b more rapidly than the simulation due to efficient secondary $\text{Cr}_{\text{Al}}^0\text{:Al}_2\text{O}_3$ PL generation by the near-edge GaN

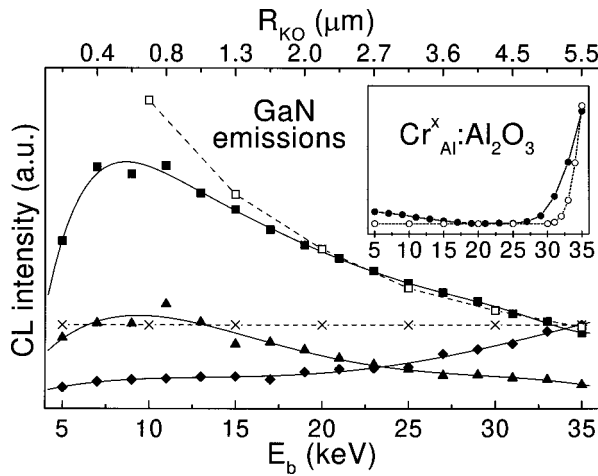


FIG. 3. Simulated and measured CL depth profiles of all emissions in the GaN/sapphire sample: ■ 3.38 eV ($\times 0.7$), ◆ YL, ▲ BL, ● $\text{Cr}_{\text{Al}}^0:\text{Al}_2\text{O}_3$, □ 3.38 eV simulation, × YL and BL simulation (constant intensity expected from a homogeneous sample since $\text{CL intensity} \propto E_b I_b$), and ○ $\text{Cr}_{\text{Al}}^0:\text{Al}_2\text{O}_3$ simulation.

emission (not accounted for by the simulation). At low beam energies, the apparent increase in the $\text{Cr}_{\text{Al}}^0:\text{Al}_2\text{O}_3$ emission with decreasing E_b is caused by a contribution to the signal by the low-energy tail of the second-order peak of the near-edge GaN emission. The near-edge emission decays exponentially with E_b (i.e., depth) due to self-absorption. The rapid decrease in the measured profile below 11 keV is caused by nonradiative surface recombination. The exponential decay with E_b implies that depth profiling of the near-edge emission in GaN layers thicker than $\sim 0.5 \mu\text{m}$ is not possible since most of the detected signal is generated within the first $0.5 \mu\text{m}$ of the epilayer at all beam energies. Consequently, interpretation of CL images acquired using the near-edge and YL or BL emissions is complicated by the differences in the generation depths of the detected signals. The effect is illustrated by the simulated CL generation depth profiles for the YL and BL emissions in Fig. 2(a) and the near-edge emission in Fig. 2(b).

The intensity of the measured YL emission increases with depth. Under constant beam power conditions, the power density decreases with increasing E_b . The increase, not present in the simulation for a homogeneous sample, therefore, suggests an increase in the YL center concentration with depth. The insignificance of YL self-absorption is illustrated by the near-edge and YL CL images (acquired at 4 K, $E_b = 15 \text{ keV}$) shown in the inset of Fig. 2(b). The YL image exhibits lower spatial resolution due to greater beam spreading since most of the signal originates deeper in the sample relative to the detected near-edge signal. We speculate that the shallow donor involved in the YL emission in the sample used in this study is O_N^+ since O was detected in the sample by qualitative WDS analysis [minimum detection

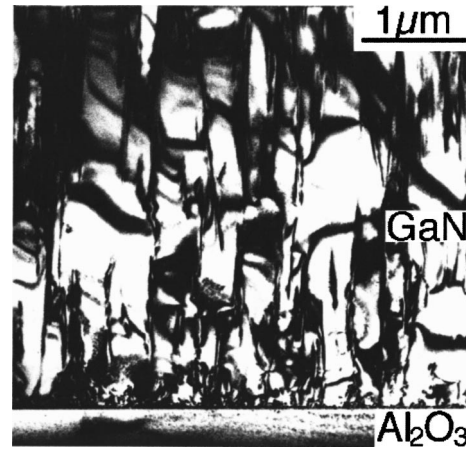


FIG. 4. Cross-sectional bright-field TEM image showing an increase in dislocation density near the GaN/sapphire interface.

limit (MDL) $\approx 500 \text{ ppm}$]. No Si was detected (MDL $\approx 600 \text{ ppm}$). An increase in O concentration with depth has been reported in depth-resolved SIMS studies of autodoped GaN grown on sapphire.^{6,7} Hence, the increase in YL intensity with depth is in agreement with the $(\text{O}_\text{N} - V_\text{Ga})^{2-}$ transition model of YL.^{2,3} An increase in V_Ga^{3-} concentration with depth is also expected due to (i) charge compensation of O_N^+ by V_Ga^{3-} during growth and (ii) an increase in the extended defect (such as dislocations) density with depth, seen in the cross-sectional TEM image of the sample (Fig. 4). The extended defects can act as traps of V_Ga^{3-} point defects.^{2,3} The BL emission intensity decreases with depth due to competitive recombination with the YL emission.

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